Salvador Balkus

Curriculum Vitae

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Research Interests

Statistical — Causal machine learning, semiparametric statistics, spatial and network data

Scientific — Environmental health, pollution, public policy

Education

2022—2027 **Doctor of Philosophy in Biostatistics**, *Harvard University*
Advised by Professor Nima Hejazi. Thesis: Nonparametric Causal Inference for Environmental Health.

2018—2022 **Bachelor of Science in Data Science**, *University of Massachusetts Dartmouth*
Summa Cum Laude. Minors: Mathematics, Economics.

Experience

2026—Present **Graduate Student Researcher**, *Massachusetts Center for Advancing Cancer Control Engaged Research Through Transformative Solutions*,
Harvard T.H. Chan School of Public Health, Boston, MA
Collaborated with clinicians and domain experts to analyze colorectal cancer screening data from community health centers (Sponsor: Prof. Karen Emmons; Investigator: Prof. Adjoa Anyane-Yeboa).

2022—Present **Graduate Student Researcher**, *NSH Lab*,
Harvard T.H. Chan School of Public Health, Boston, MA
Designed and implemented computational advances in statistical learning and causal inference methodology, including open-source software in R and Julia (Advisor: Prof. Nima Hejazi).

2022—Present **Graduate Student Researcher**, *National Studies on Air Pollution and Health*,
Harvard University, Cambridge, MA
Developed new causal inference methods for spatial and network data with continuous treatments; initially through the Training Program in Environmental Health Statistics, T32 ES007142
(Lab PI: Prof. Francesca Dominici; Mentor: Prof. Rachel Nethery).

2020—2022 **Research Assistant**, *Computational Statistics & Data Science Lab*,
University of Massachusetts Dartmouth, North Dartmouth, MA
Implemented algorithms for smart health and dietary data analysis (Advisor: Prof. Julia Hua Fang).

Research Fellowships

2022 **Graduate Research Fellowship**, *National Science Foundation*

Awards

2025 **Early Career Travel Award**, *Joint Statistical Meetings*, Nashville, TN

2024—2025 **Certificate of Distinction in Teaching**, *BST 232: Methods I*, Harvard University

2022 **Best Analysis and Visualization**, *SEMASS DataFest*, American Statistical Association

2021 **Travel Award**, *Joint Mathematics Meetings*, American Mathematical Society

Publications and Working Papers

Statistical Methodology

[1] **S. V. Balkus**, C. Testa, and N. S. Hejazi, "A Riesz representer perspective on Targeted Learning", *Accepted with revisions to the International Journal of Biostatistics*, 2026.

[2] **S. V. Balkus**, S. W. Delaney, and N. S. Hejazi, "The causal effects of modified treatment policies under network interference", *Journal of the Royal Statistical Society Series B: Statistical Methodology*, vol. 88, no. 4, pp. 1407–1426, Mar. 2026, ISSN: 1467-9868. DOI: 10.1093/jrsssb/qkag052.

[3] **S. V. Balkus** and N. S. Hejazi, "CausalTables.jl: Simulating and storing data for statistical causal inference in Julia", *Journal of Open Source Software*, vol. 10, no. 106, p. 7580, 2025. DOI: 10.21105/joss.07580.

Preprints

[4] E. Tong and **S. V. Balkus**, "Linear models for causal inference under network interference", *arXiv preprint arXiv:2603.29168*, 2026.

[5] **S. V. Balkus**, H. Laith, and N. S. Hejazi, "On the use of cross-fitting in causal machine learning with correlated units", *arXiv preprint arXiv:2601.10899*, 2026.

Works in Preparation

[6] **S. V. Balkus** and N. S. Hejazi, "Quickly approximating a learning algorithm while preserving its statistical rate", *In preparation*, 2026.

[7] **S. V. Balkus** and N. S. Hejazi, "Function estimation under network interference", *In preparation*, 2026.

[8] C. Jiang, **S. V. Balkus**, and N. S. Hejazi, "A unified projection theory for conditional and coarsened mean functionals", *In preparation*, 2026.

Applied Computer Science

[9] **S. V. Balkus** and D. Yan, "Improving short text classification with augmented data using GPT-3", *Natural Language Engineering*, pp. 1–30, 2023. DOI: 10.1017/s1351324923000438.

[10] **S. V. Balkus**, H. Fang, and H. Wang, "Federated fuzzy clustering for longitudinal health data", *2022 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE)*, pp. 128–132, 2022.

[11] **S. V. Balkus**, H. Wang, B. D. Cornet, C. Mahabal, H. Ngo, and H. Fang, "A survey of collaborative machine learning using 5G vehicular communications", *IEEE Communications Surveys & Tutorials*, vol. 24, no. 2, pp. 1280–1303, 2022. DOI: 10.1109/comst.2022.3149714.

[12] V. S. Gurugubelli, H. Fang, J. M. Shikany, **S. V. Balkus**, J. Rumbut, H. Ngo, H. Wang, J. J. Allison, and L. M. Steffen, "A review of harmonization methods for studying dietary patterns", *Smart Health*, vol. 23, p. 100 263, 2022. DOI: 10.1016/j.smhl.2021.100263.

[13] **S. V. Balkus**, H. Fang, J. Rumbut, A. Moormann, and E. Boyer, "A multi-level biosensor-based epidemic simulation model for COVID-19", *IEEE Internet of Things Journal*, pp. 1–1, 2021. DOI: 10.1109/jiot.2021.3127804.

[14] **S. V. Balkus**, J. Rumbut, H. Wang, and H. Fang, "An adaptive and dynamic biosensor epidemic model for COVID-19", in *2020 IEEE 21st International Conference on Information Reuse and Integration for Data Science (IRI)*, IEEE, Aug. 2020. DOI: 10.1109/iri49571.2020.00051.

Presentations

Conferences

[1] **S. V. Balkus**, H. Laith, and N. S. Hejazi, "Cross-fitting in causal machine learning with correlated units", in *Joint Statistical Meetings*, Aug. 2026.

[2] **S. V. Balkus**, C. Testa, and N. S. Hejazi, "A Riesz representer perspective on Targeted Learning", in *European Causal Inference Meeting*, Apr. 2026.

[3] **S. V. Balkus** and N. S. Hejazi, "Lasso rates hold even after randomly discarding most of the predictors", in *Joint Statistical Meetings*, Aug. 2025.

[4] **S. V. Balkus**, R. C. Nethery, S. W. Delaney, and N. S. Hejazi, "Nonparametric network causal inference for continuous exposures in mobile source air pollution", in *American Causal Inference Conference*, May 2024.

[5] **S. V. Balkus**, R. C. Nethery, S. W. Delaney, and N. S. Hejazi, "Assumption-lean causal inference for mobile source air pollution", in *ASA Boston Chapter Student Research Symposium on Statistics and Data Science*, Apr. 2024.

[6] **S. V. Balkus** and D. Yan, "Language models that teach themselves: Augmenting training data for topic classification using GPT-3", in *ASA Boston Chapter Student Research Symposium on Statistics and Data Science*, Apr. 2022.

[7] N. Pai, **S. V. Balkus**, and T. Zeng, "Multi-hypothesis tracking of space objects and targets", in *AMS Joint Mathematics Meetings (JMM) Poster Session*, Apr. 2022.

[8] **S. V. Balkus**, J. Rumbut, and H. Fang, "Multi-level biosensor-based epidemic forecasting in small areas", in *Joint Statistical Meetings*, American Statistical Association, Aug. 2021.

Invited Talks

[9] **S. V. Balkus**, *Remarks as Discussant for "Clinical decision support for HIV care: development, implementation, evaluation"*, Harvard Data Science Initiative Causal Seminar, Oct. 2026.

[10] **S. V. Balkus**, C. Testa, and N. S. Hejazi, "TMLE made easy with Riesz representers", in *Harvard Biostatistics Student Seminar*, Sep. 2026.

[11] **S. V. Balkus**, S. W. Delaney, and N. S. Hejazi, "Induced modified treatment policies for estimating effects of continuous exposures under interference", in *University of Washington TGIF Reading Group*, May 2026.

[12] S. Woodward and **S. V. Balkus**, "Spatial confounding and interference in environmental health studies", *BUSPH-HSPH CAFE Research Coordinating Center Webinar*, Feb. 2026.

[13] **S. V. Balkus**, S. W. Delaney, and N. S. Hejazi, "The causal effects of modified treatment policies under network interference", in *Space-Time Causality Reading Group*, `https://spacetimecausality.github.io/`, Sep. 2025.

[14] **S. V. Balkus**, "Speeding up the Highly Adaptive Lasso", in *Harvard Biostatistics Student Seminar*, Mar. 2025.

[15] **S. V. Balkus**, "Statistics in Julia: Is it right for you?", in *Harvard Biostatistics Student Seminar*, Dec. 2024.

[16] **S. V. Balkus**, "Lunchtime Computing: Basics of AWS SageMaker", in *Center for Scientific Computing and Visualization Research*, University of Massachusetts Dartmouth, Feb. 2021.

[17] **S. V. Balkus**, "Lunchtime Computing: Getting started with Git and GitHub", in *Center for Scientific Computing and Visualization Research*, University of Massachusetts Dartmouth, Feb. 2020.

Mentoring

2026—Present **NSH Lab**, *Jolene Cao*, Project: "Simple Riesz learners for causal inference"

2025—2026 **NSH Lab**, *Hasan Laith*, Project: "On the use of cross-fitting for causal machine learning in dependent-data settings"

2025—2026 **National Studies on Air Pollution and Health**, *Eric Tong*, Project: "Linear models under network interference"

Summer 2025 **Harvard StatStart**, *Maria Huang, Moulik Pagarani, and Ian Ruben Rivera Rodriguez*, Project: "Mapping the Legacy of Redlining"

Teaching

Fellowships

Fall 2026 **Teaching Fellow**, *BST 220: Applied Generalized Linear Models*, Harvard University
For master's students in biostatistics and across fields of public health to learn practical aspects of linear modeling; held office hours, graded, and taught labs.

Fall 2024 **Teaching Fellow**, *BST 232: Methods I*, Harvard University
For PhD and master's students to learn theory of linear modeling, hypothesis testing, computational methods, and Bayesian statistics; held office hours, graded, and taught & designed labs.

Summer 2024 **Curriculum Fellow**, *BST 232: Methods I*, Harvard University
Aided in curriculum redesign to split Methods into two courses; created new, improved labs and instructional materials for Methods I.

Fall 2023 **Teaching Fellow**, *BST 232: Methods*, Harvard University
For PhD and master's students to learn theory of linear models, computational methods, categorical data analysis, and generalized linear models; held office hours, graded, and taught & redesigned labs.

Programs

2024—2025 **Qualifying Examination Tutor**, Harvard University
Taught refresher on regression and optimization topics and led review sessions for first-year Biostatistics PhD Qualifying Examination.

July 2025 **Research Project Instructor**, *Harvard StatStart*, Harvard University
For underserved high-school students in the Boston area to be introduced to statistics; designed month-long high-school-level data analysis project in R and mentored research team of three.

June 2024 **Workshop Instructor**, *Replicathon, IQ BIO REU*, University of Puerto Rico Río Piedras
Co-led 2-day, in-person workshop in Puerto Rico for undergraduate students in the biological sciences on data analysis in R.

Service and Extracurricular

Ad-hoc Reviewer
Journal of the Royal Statistical Society Series B (Statistical Methodology), Journal of Causal Inference [3x], The R Journal, Journal of Open Source Software [4x], Natural Language Processing, BMC Medical Informatics and Decision Making

2024 —Present **Co-Chair**, *Biostatistics Student Committee*, Harvard University
Led peer mentoring program, qualifying exam tutoring program, social event planning, and resource organization for 70+ PhD students.

2022 —2026 **Graphic Designer and Blog Contributor**, *Science in the News*, Harvard University
Designed infographics and wrote blog posts explaining scientific topics to a lay audience.

2025 **Session Chair, Causal Discovery**, *Joint Statistical Meetings*, Nashville, TN

2023 —2024 **Committee Chair**, *Biostatistics Peer Mentoring Program*, Harvard University
Supported class of 12 incoming students by serving as peer mentor, creating instructional material, acquiring event funding, and planning seminar series.

2022 **Session Chair, Biostatistics**, *Student Research Symposium on Statistics and Data Science*, ASA Boston Chapter

Scientific Blog Contributions

[1] H. Farnsworth (Author) and **S. V. Balkus (Illustrator)**, "Take a deep breath: How stress reduction may influence cancer risk", *Harvard Science in the News*, Jul. 2026.

[2] E. Crowley-Dolen (Author) and **S. V. Balkus (Illustrator)**, "Suzetrigine and the shift toward non-opioid pain management", *Harvard Science in the News*, Oct. 2025.

[3] A. Bogatch (Author) and **S. V. Balkus (Illustrator)**, "Pregnancy causes astonishing brain changes — why haven't we noticed this before?", *Harvard Science in the News*, May 2025.

[4] S. Renauld (Author) and **S. V. Balkus (Illustrator)**, "Gene editing to treat disease: A success story in sickle cell disease", *Harvard Science in the News*, May 2024.

[5] H. Blume (Author) and **S. V. Balkus (Illustrator)**, "Liquid-within-liquid compartments: A groundbreaking discovery in biology", *Harvard Science in the News*, Jun. 2023.

[6] **S. V. Balkus (Author and Illustrator)**, "How do scientists know whether to trust their results?", *Harvard Science in the News*, Jan. 2023.

Skills

Languages `R`, `Julia`, `Python`, `Java`, `SQL`, LaTeX, markdown, `HTML/CSS`

Tools Git, GitHub, Quarto, RStudio, VSCode, AWS, SLURM, Linux

Open-Source Contributor Credits

- The Book of Statistical Proofs, `https://statproofbook.github.io/`
- TMLE.jl, `https://github.com/TARGENE/TMLE.jl`
- DensityRatioEstimation.jl, `https://github.com/JuliaML/DensityRatioEstimation.jl`
- `{nadir}`, `https://github.com/ctesta01/nadir`
- `{RieszCML}`, `https://github.com/nshlab/RieszCML`

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